Andrew Mueller

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GitHub
Google Scholar

EDUCATION

Doctor of Philosophy - California Institute of Technology, Pasadena — Defense Date: Dec 11th, 2023
Option: Applied Physics — Graduation: June 2024
Thesis: Quantum Measurements with SNSPDs
Caltech Advisor: Maria Spiropulu
JPL Advisor: Matthew Shaw

Bachelor of Science - University of California, San Diego
Major: Physics — Graduation: March 2017
Minor: Engineering Mechanics — GPA: 3.803

AWARDS AND FELLOWSHIPS

Phi Beta Kappa Society (May 2015)
Rose Hills Fellowship (Sept 2017)
Brinson Foundation Fellow (Sept 2020)

EXPERIENCE & PUBLICATIONS

Postdoctoral Scholar, JPL Microdevices Group, SNSPD subgroup — Jan 2026 – Present
Projects:

- Research and development of high-density interconnects for SNSPD readout
  - Study cable heat load, crosstalk, model attenuation
  - High density polyimide stripline flex cable modelling
- Cabling test & measurement for DARPA-funded microwire array cryostat
- Lab-wide rewrite of measurement software emphasizing remote access and AI-assisted automation patterns.
- Automation and remote control of photon count rate measurements in a mid-IR SNSPD measurement testbed
- Co-corresponding author on 2 forthcoming mid-IR SNSPD papers

Postdoctoral Scholar, INQNET under Prof. Maria Spiropulu, Caltech — March 2024 – Dec. 2025
Projects:

- Test & measurement of mid-infrared SNSPDs for dark matter searches
  - Hardware and software development of a cryogenic RF switch. Increased measurement velocity for mid-IR SNSPDs by 8x
  - Design and testing of a light-tight SNSPD measurement package with removable Leadless Chip Carrier (LCC), dramatically decreasing wire-bonding and device installation time.
- Modelling of cavity-enhanced waveguide coupled ppKTP pair generating crystal for multiplexed quantum networks
- Installation and testing of near-infrared differential single pixel SNSPDs, manufacture of amplifier readout boxes
- Software and hardware development for the Device Bay electronics platform

Graduate Student, JPL Microdevices Laboratory; INQNET program Sept 2018 – Dec. 2023

- Non-published projects
  - Packaging, building, and testing of low jitter photon-number-resolving 1550nm SNSPDs for quantum networks. Fiber splicing, device packaging, building readout electronics

Recent Publications (2020 – Present):

A. Mueller, S. I. Davis, B. Korzh, R. Valivarthi, A. D. Beyer, R. Youssef, N. Sinclair, C. Peña, M. D. Shaw, and M. Spiropulu, "High-rate multiplexed entanglement source based on time-bin qubits for advanced quantum networks," *Optica Quantum* **2**, 64–71 (2024).

A. Mueller, E. E. Wollman, B. Korzh, A. D. Beyer, L. Narvaez, R. Rogalin, M. Spiropulu, and M. D. Shaw, "Time-walk and jitter correction in snspds at high count rates," Physics Letters 122, 044001 (2023).

A. S. Mueller, B. Korzh, M. Runyan, E. E. Wollman, A. D. Beyer, J. P. Allmaras, A. E. Velasco, I Craiciu, B. Bumble, R. M. Briggs, L. Narvaez, C. Peña, M. Spiropulu, and M. D. Shaw, "Free-space coupled superconducting nanowire single-photon detector with low dark counts," Optica 8, 1586–1587 (2021).

S. I. Davis, **A. Mueller**, R. Valivarthi, N. Lauk, L. Narvaez, B. Korzh, A. D. Beyer, O. Cerri, M. Colangelo, K. K. Berggren, M. D. Shaw, S. Xie, N. Sinclair, and M. Spiropulu, "Improved heralded single-photon source with a photon-number-resolving superconducting nanowire detector," Rev. Appl. 18, 064007 (2022).

R. Valivarthi, S. I. Davis, C. Peña, S. Xie, N. Lauk, L. Narváez, J. P. Allmaras, A. D. Beyer, Y. Gim, M. Hussein, et al., "Teleportation systems toward a quantum internet," PRX Quantum 1, 020317 (2020).

R. Valivarthi, L. Narvarez, S. I. Davis, N. Lauk, C. Peña, S. Xie, J. P. Allmaras, A. D. Beyer, B. Korzh, **A. Mueller**, M. Kiburg, M. D. Shaw, E. E. Wollman, P. Spentzouris, D. Oblak, N. Sinclair, and M. Spiropulu, "Picosecond synchronization system for quantum networks," Journal of Lightwave Technology , 1–7 (2022).

See more publications and conference proceedings on google scholar.

Engineer, Basov Infrared Laboratory Dec 2013 – June 2017
University of California San Diego (lab since moved to Columbia University)

- Design of Terahertz optical system with parabolic mirrors using ZEMAX EE
- Conceptual design of new Cryogenic Infrared Atomic Force Microscope

SKILLS

- Full-stack web development with Typescript, Svelte, Tailwind, Fastapi, MongoDB, Docker, Rust & WebAssembly. CI/CD experience. AI-assisted workflows
- Proficient with Python, Numpy, PyTorch, C, Matlab, Wolfram Mathematica
- Proficient with Solidworks for research/prototyping, ZEMAX for optical design
- 3d and 2d Graphics, rendering. Creator of Nature cover artwork, and press-release artwork